

Cyrus Guided Notes: Stanford CS224N Natural Language Processing with Deep Learning

Stanford course pack

2026-05-15

Course source

Official source: <https://web.stanford.edu/class/cs224n/>

Why this course is in the system. Language and representation learning for Hermes interaction, tutoring, and multimodal reasoning.

Learning output. Keep it as the NLP and Transformer side line after the spatial geometry first round.

Prerequisite checkpoint

- Linear algebra
- Probability
- Neural networks

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned}\text{sim}(\mathbf{u}, \mathbf{v}) &= \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \\ \text{Attention}(Q, K, V) &= \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \\ \mathbf{h}_t &= f(\mathbf{x}_t, \mathbf{h}_{t-1}) \\ L &= - \sum_t \log p(y_t | y_{<t}, \mathbf{x})\end{aligned}$$

Visual page

Show tokens -> embeddings -> attention map -> context vector -> prediction.

GoodNotes pages

- Page ST-NLP001: embeddings
- Page ST-NLP002: attention
- Page ST-NLP003: sequence loss

Web interaction

An attention heatmap that highlights which token reads which token.

Obsidian and Notion

Layer	Output
Obsidian	Stanford NLP -> CS224N
Notion	Hermes interaction bridge and assignment status.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.